

NeuralPathReasoning-MIND

Neural Bellman-Ford Networks for Drug Repurposing | Project Report - June 2026

1. Project Overview

NeuralPathReasoning-MIND is a follow-on project to WeightedKgBlend, exploring whether state-of-the-art neural path reasoning methods - specifically the ULTRA model based on Neural Bellman-Ford Networks (NBFNet) - can dramatically improve drug repurposing performance on the MIND biomedical knowledge graph.

While WeightedKgBlend achieved its best result with Path-Gated Re-ranking (MRR = 0.0682), this approach was limited by ProbCBR path coverage (~41%) and shallow 1-2 hop reasoning. ULTRA applies a generalised Bellman-Ford algorithm over graph neural networks, learning to propagate and combine relational paths of arbitrary length in a fully differentiable way - with no task-specific fine-tuning required.

2. The Algorithm: Neural Bellman-Ford Networks (NBFNet / ULTRA)

2.1 What is NBFNet?

NBFNet (Zhu et al., NeurIPS 2021) reformulates link prediction as a generalised Bellman-Ford shortest-path problem on knowledge graphs. For a query (drug, treats, ?), it computes a "path representation" from the source node (drug) to every candidate target node (disease) by iteratively aggregating multi-hop relational paths through the graph using learned message functions.

Key properties of NBFNet:

- Differentiable: the Bellman-Ford iterations are implemented as graph neural network layers, making the whole pipeline end-to-end trainable.
- Path-aware: unlike TransE/RotatE which score triples independently, NBFNet explicitly models multi-hop reasoning paths between entities.
- Inductive: representations are computed at inference time, allowing generalisation to unseen entities.

2.2 What is ULTRA?

ULTRA (Galkin et al., NeurIPS 2023) extends NBFNet to a universal, transferable foundation model for knowledge graph reasoning. It is pre-trained on 50 diverse knowledge graphs simultaneously, learning relation representations that transfer zero-shot to new graphs - including biomedical KGs like MIND that were never seen during training.

ULTRA advantages over WeightedKgBlend approaches:

- Zero-shot transfer: no training on MIND required - ULTRA applies directly.
- Arbitrary-depth paths: Bellman-Ford iterations propagate through the full graph, not limited to 1-2 hop paths like ProbCBR.
- All predictions have implicit path support: every score is derived from relational path aggregation, giving biological interpretability by design.
- Pre-trained model: `mgalkin/ultra\_50g` available on HuggingFace.

3. What We Have Done

3.1 Dataset

We evaluate on MIND splits_updated - the same updated dataset used in WeightedKgBlend - comprising 6,761 FDA-approved drug-disease indication edges across 5 stratified cross-validation folds (676 test / 676 valid / ~5,409 train per fold). This ensures direct comparability with all WeightedKgBlend baselines.

3.2 Zero-Shot ULTRA Inference (Completed)

We ran ULTRA (ultra_50g, pre-trained on 50 graphs) in zero-shot mode on all 5 folds of MIND splits_updated using a Kaggle GPU notebook. No fine-tuning was performed - ULTRA was applied directly to MIND.

Per-fold test results:

Fold	MRR	Hits@1	Hits@3	Hits@5	Hits@10
Slice 0	0.3666	0.2845	0.3851	0.4315	0.4955
Slice 1	0.3469	0.2704	0.3676	0.4168	0.4955
Slice 2	0.3691	0.2918	0.3914	0.4360	0.5305
Slice 3	0.3532	0.2782	0.3738	0.4212	0.4888
Slice 4	0.3486	0.2870	0.3716	0.4085	0.4917
Mean +/- std	0.3569 +/- 0.009	0.2824 +/- 0.010	0.3779 +/- 0.009	0.4232 +/- 0.009	0.5004 +/- 0.015

3.3 Comparison to WeightedKgBlend (test set)

Method	MRR	Hits@1	Hits@10
TransE	0.0158	0.0036	0.0311
ProbCBR	0.0138	0.0050	0.0296
RotatE	0.0460	0.0135	0.1100
Simple Ensemble	0.0489	0.0135	0.1097
Path-Gated Re-ranking	0.0682	0.0188	0.1291
ULTRA zero-shot	0.3569	0.2824	0.5004

ULTRA zero-shot outperforms Path-Gated Re-ranking by 5.2x in MRR and 3.9x in Hits@10. Hits@10 = 0.50 means the correct disease appears in the top-10 predictions for half of all drug queries, with no training on MIND whatsoever.

4. Future Plan

4.1 Fine-Tune ULTRA on MIND (Next Step)

Zero-shot ULTRA is already a strong baseline. Fine-tuning on MIND splits_updated is expected to push performance substantially further. The plan:

- Use ULTRA fine-tuning API (GraFN / ULTRA fine-tune script from the authors) on Kaggle GPU (P100/T4).
- Train for 10-20 epochs on each fold's training set (5-fold CV, same splits as WeightedKgBlend for direct comparison).
- Evaluate with filtered MRR / Hits@K on test and valid sets.
- Target: MRR > 0.45 (expected gain of ~25-30% over zero-shot based on reported fine-tuning gains in the ULTRA paper).

4.2 Mechanistic Path Extraction from NBFNet

A key advantage of NBFNet over simple KGE methods is that each prediction score is computed via Bellman-Ford path aggregation - meaning interpretable paths can be extracted. The plan:

- Use attention weights from NBFNet's final layer to identify the highest-contributing relational path for each drug-disease prediction.
- Extract top-1 mechanistic path for each of the 455 WeightedKgBlend repurposing candidates and compare to ProbCBR paths.
- This provides a richer, deeper mechanistic explanation than ProbCBR's 1-2 hop paths, potentially uncovering novel multi-step biological mechanisms.

4.3 Drug Repurposing Candidate Re-analysis

Re-run the 455 high-confidence candidate extraction pipeline using ULTRA fine-tuned predictions instead of Path-Gated predictions:

- Filter novel drug-disease pairs not in any training/valid/test indication set.
- Rank by ULTRA score, extract NBFNet mechanistic paths.
- Curate top candidates with biological plausibility assessment.
- Compare candidate lists between ULTRA and Path-Gated to identify where the two methods agree (high-confidence) vs. diverge.

4.4 Integration with WeightedKgBlend Paper

Results from this project will feed into a follow-up paper or extended version of the WeightedKgBlend manuscript:

- Add ULTRA as a new baseline/method in the results tables.
- Position the work as a three-tier progression: simple KGE -> Path-Gated (hybrid) -> ULTRA (neural path reasoning).
- Discuss the trade-off: ULTRA is more powerful but less transparent than ProbCBR paths; ULTRA + NBFNet path extraction bridges this gap.

4.5 Longer-Term

- Explore NBFNet trained from scratch on MIND (vs. ULTRA transfer).
- Investigate inductive setting: predict for drugs/diseases not seen in training.
- Extend to multi-relational queries beyond drug-disease indication (e.g., drug-target, disease-gene).

5. Summary

Project: NeuralPathReasoning-MIND

Algorithm:	ULTRA / NBFNet - Neural Bellman-Ford Networks for KG reasoning
Dataset:	MIND splits_updated - 6,761 indication edges, 5-fold CV
Completed:	Zero-shot ULTRA inference on all 5 folds (MRR = 0.357 +/- 0.009)
Key result:	5.2x MRR improvement over Path-Gated Re-ranking, zero training needed
Next step:	Fine-tune ULTRA on MIND + extract NBFNet mechanistic paths
Goal:	New state-of-the-art drug repurposing with interpretable deep paths

6. References

- [1] Zhu, Z. et al. (2021). Neural Bellman-Ford Networks: A General Graph Neural Network Framework for Link Prediction. NeurIPS 2021.
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- [4] Das, R. et al. (2020). Probabilistic Case-Based Reasoning for Open-World Knowledge Graph Completion. EMNLP 2020.
- [5] Sinha, M. et al. (2025). Combining Knowledge Graph Embeddings and Mechanistic Paths for Drug Repurposing. bioRxiv (WeightedKgBlend).